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Welcome to CodeSport! In this month's column, we feature a medley of questions covering C/C++, data structures, algorithms and operating systems.

We close the year by featuring a bunch of questions based on the various topics we discussed in this column over the last 12 months. These questions require you to apply some of the concepts we covered through 2010 and should help readers in preparing for software interviews. Let's get started with some easy questions first.

1. Given a stack data structure that allows these operations: *push*, *pop*, *is_empty*, and *peek* (returns the value of the topmost element without popping it), you are asked to write a code fragment to sort the elements of the stack in descending order. In other words, once the stack is sorted, the stack top should contain the maximum element, the element next to the top should be the second maximum and so on. Assume that the stack contains N integers. What is the complexity of your algorithm?
2. Given a directed graph $G(V, E)$ where V is the set of vertices and E is the set of edges, you are asked to find if there exists a path between two given vertices (x, y) . What would be the complexity of your solution? If you are given a set of vertex pairs and asked to find if there exists a path for each pair of vertices in the set, what would your approach to the problem be?
3. Given a DFS representation of a binary tree, with each vertex and its associated DFS number represented as (v, n) where v is the vertex ID and n is the corresponding DFS number of the vertex v , write an algorithm to give a BFS representation of the same binary tree. Assume that we number the root of the tree with the DFS number 1.
4. Given an undirected graph $G(V, E)$ where V is the set of vertices and E is the set of edges, give an algorithm to determine whether the graph is 2-colourable. (By 2-colourable, we mean that the graph vertices can be coloured with 2 colours such that no two adjacent vertices have the same colour.)
5. Given a set S of N integers, write an algorithm to determine all the subsets of S . What is the complexity of your solution? If you are asked to find all the k -subsets (subsets with k members), what would your approach be?
6. Given a two-dimensional matrix ($m \times n$) of integer elements, in which each row of the matrix and each column of the matrix is already sorted, give an algorithm to find if a number x is present in the matrix.
7. Write a small code snippet to find whether the system stack grows upward or downward on your system.
8. Given a set of cities, and the distances between the cities, you are asked to visit all the cities as part of a tour. Write a program which will ensure that you visit all cities, but travel the minimum number of miles overall in the course of your journey. What is the complexity of your solution?
9. Given an undirected graph $G(V, E)$ where V is the set of vertices and E is the set of edges, write a program to determine whether it is a tree.
10. In C++ programs, when would you prefer to declare your class' constructor, copy constructor and assignment operator as private functions?
11. We know that *malloc* and *free* are standard library functions for allocating and deallocating memory dynamically in C programs. When we use *free*, we only pass the pointer to the memory to be deallocated as a parameter, and not the size of the memory block to be deallocated. How does the *free* library call determine what size the block to be deallocated should be? Give your implementations for *malloc* and *free* with an explanation.